

Treat the following exercises:

I. (4 pts)

Here is a DTD for a musical library:

```
<! ELEMENT cd (title, artist+, song*)>
<! ELEMENT title (#PCDATA)>
<! ELEMENT artist ((firstname*, name) | group ?)>
<! ELEMENT firstname (#PCDATA)>
<! ELEMENT name (#PCDATA)>
<! ELEMENT group (#PCDATA)>
<! ELEMENT song (#PCDATA)>
<! ATTLIST song year CDATA #IMPLIED>
<! ATTLIST song kind (A1 | B1) "B1">
<! ATTLIST song duration CDATA #REQUIRED>
```

Create a well-formed and valid XML document for this DTD.

II. (4 Pts)

Write an internal DTD for the following XML file:

```
<?xml version= "1.0"?>
<cd>
<title>Beautiful Illness</title>
<author>
<name>Waits</name>
<firstname>Tom</firstname>
<address>US</address>
</author>
<song kind="urban folk" year="2000"/>
</cd>
```

III. (4 pts)

Given the following Access table (persons):

Name	First name	Address	e-mail	telephone
N1	P1	A1	e1@email.com	4167894
N2	P2	A2	e2@email.com	4167235

Convert this table into a XML file, we consider e-mail and telephone as attributes.

IV. (3 pts)

Given the following XML file:

```
<A>
<B>
<E>4</E>
<E>3</E>
</B>
<B>
<E>1</E>
</B>
<C>
<F>a</F>
</C>
<E>2</E>
</A>
```

Give the results of the following XPATH expressions:

- /A/B
- /A/B[1]

V. (5 pts)

Given the following XML file:

```
<?xml version="1.0" ?>
<catalog>
  <cd>
    <title>Hide your heart</title>
    <artist>Bonnie Tyler</artist>
    <price>9</price>
  </cd>
  <cd>
    <title>Mother</title>
    <artist>Dr. Bee Gees</artist>
    <price>15</price>
  </cd>
</catalog>
```

a) Write a XSL file to convert this file under the following form:

Title	Artist
Hide your heart	Bonnie Tyler
Mother	Dr. Bee Gees

b) Modify the XSL file to only display the titles and the artists that have a price>10.